

Isha Chaudhary

ishachaudhary.web.illinois.edu

isha4@illinois.edu

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Education

University of Illinois Urbana-Champaign

2022 - 2027

Ph.D. in Computer Science; Guide: Prof. Gagandeep Singh

Committee: Prof. Sasa Misailovic, Prof. Madhusudan Parthasarathy, Prof. Clark Barrett

University of Illinois Urbana-Champaign

2022 - 2025

M.S. in Computer Science; Guide: Prof. Gagandeep Singh

GPA: 3.97/4.00

Thesis: Assessing Trustworthiness of Neural Networks for Computer Systems

Indian Institute of Technology Delhi, India

2018 - 2022

B.Tech in Electrical Engineering (minor in CS); GPA: 9.3/10.0

Class Rank: 1/51

Research Interests

Trustworthy Frontier models, Neural Networks for Computer Systems

Selected Publications

- **Isha Chaudhary**, Vedaant Jain, Avaljot Singh, Kavya Sachdeva, Sayan Ranu, Gagandeep Singh. *Lumos: Let there be Language Model System Certification*. Arxiv 2025.
- **Isha Chaudhary**, Qian Hu, Manoj Kumar, Morteza Ziyadi, Rahul Gupta, Gagandeep Singh. *Quantitative Certification of Bias in Large Language Models*. **ICLR 2025**.
- **Isha Chaudhary**, Vedaant Jain, Gagandeep Singh. *QuaCer-C: Quantitative Certification of Knowledge Comprehension in LLMs*. **AISTATS 2026 Spotlight**, also in SeT LLM Workshop @ ICLR 2024.
- Chengxiao Wang, **Isha Chaudhary**, Qian Hu, Weitong Ruan, Rahul Gupta, Gagandeep Singh. *Quantifying Risks in Multi-turn Conversation with Large Language Models*. **ICLR 2026**.
- **Isha Chaudhary**, Alex Renda, Charith Mendis, Gagandeep Singh. *COMET: X86 Cost Model Explanation Framework*. **MLSys 2024**, also in XAI-in-Action Workshop @ NeurIPS 2023.
- **Isha Chaudhary**, Shuyi Lin, Cheng Tan, Gagandeep Singh. *Specification Generation for Neural Networks in Systems*. **ICSE 2026** (Poster), also in ML4Wireless Workshop @ ICML 2025.
- Jason Vega*, **Isha Chaudhary***, Changming Xu*, Gagandeep Singh. *Bypassing the Safety Training of Open-Source LLMs with Priming Attacks*. Tiny Papers @ ICLR 2024. (* indicates equal contribution)

Experience

Amazon AGI

Boston, USA

Applied Scientist Intern with Dr. Rahul Gupta and Dr. Weitong Ruan

May 2026 - Aug 2026

Amazon Web Services

Seattle, USA

Applied Scientist Intern with Dr. Ganyu Teng and Dr. Shawn Zhang

Sep 2025 - Dec 2025

Finetuned Qwen-1.5B with GRPO to generate prompt-injection attacks against scalable-oversight in agentic systems.

Microsoft Research

Redmond, USA

Research Intern with Dr. Ryan Beckett, Dr. Siva Kakarla, Prof. Francis Yan,

Dr. Behnaz Arzani

May 2025 - Aug 2025

Developed novel multivariate anomaly detection methods over system logs that leverage log semantics using LLMs.

Microsoft AI (Bing)

Redmond, USA

Data Science Intern

May 2024 - Aug 2024

Developed machine learning models for revenue optimization in Bing search.

Adobe Research (Cloud Tech group)

Bengaluru, India

Research Intern / Guide: Dr. Gaurav Sinha / US patent no. US 2023/0274310 A1

May 2021 - Aug 2021

Multi-task Learning with Label Proportions: Jointly disaggregated multiple aggregate labels with individual features

Blockchain IoT Security Project: DST-JST Grant on IoT Security

IIT Delhi, India

Research Assistant / Guide: Prof. Subodh Sharma, IIT Delhi / Denso Intl. India

Sept 2020 - May 2021

Designed & developed Hyperledger Fabric-based supply chain system for real-time order traceability & security

Awards

- Awarded **Best Presentation Award** (Security and Privacy track) at CSL student conference 2025 (2025)
- Awarded **Institute Silver Medal & Prof. C.S. Jha Memorial Excellence Award** at IIT Delhi (2022)
- **Top 7% merit award**: In top 7% high GPA students at IIT Delhi for 5 semesters: semesters 3, 4, 6, 7, 8 (2019-22)
- **Design & Innovation Summer Award (MHRD Grant, Govt. of India)** (2019)
- **KVPY**: Qualified for the Kishore Vaigyanik Protsahan Yojana Fellowship (IISc); All India Rank 232 (2016)
- **NTSE**: Received National Talent Search Scholarship, given to top 1000 students nationwide by NCERT (2015)

Talks

Certifying LLMs

- National AI Research Resource (NAIRR) Pilot 2026
- Microsoft Research Redmond and Microsoft Research India 2025
- CSL student conference 2025
- IIT Delhi 2025
- Amazon-Illinois Center on AI for Interactive Conversational Experiences 2024

Teaching Smarter with AI: How to Use AI in Classrooms Without Losing Control

- Graduate TA training academy organized by CITL, UIUC 2025

Exploring COMET: Neural Cost Model Explanation Framework

- Google Compiler Research reading group 2024
- VMWare Research 2024

Teaching

Grad Academy, CITL

UIUC, USA

Instructor: Isha Chaudhary, UIUC

Aug 19, 2025

Advanced Topics in Programming Systems - Trustworthy AI Systems (CS521)

UIUC, USA

Instructor: Prof. Gagandeep Singh, UIUC

Aug 2024 – Dec 2024

Calculus (MTL100)

IIT Delhi, India

Instructor: Prof. Sivananthan Sampath, IIT Delhi

Aug 2021 – Dec 2021

Grants

- Awarded **ICML 2023 Travel Grant** (2023)
- Awarded scholarship for **Programming Languages Mentoring Workshop @ PLDI 2023** (2023)
- **NSF Travel Grant for IEEE SaTML 2023** (2022)

Service

- Area Chair for MLSys 2025.
- Reviewer for ICML 2026; ICLR 2025, 2026; FAccT 2025, 2026; MLSys 2025; AIES 2024, 2025; Tiny Papers @ ICLR 2024.
- Artifact Evaluator for VMCAI 2024.
- Main organizer for Formal Methods and Software Engineering seminar, Spring 2024 & Fall 2024, at UIUC.
- Mentor for the Undergraduate Research Apprenticeship Program (URAP) at UIUC, 2024-25
- Mentor in the Undergraduate Research in Scientific Advancement (URSA) program at UIUC, 2024-25
- Mentor in SIGPLAN-M long-term mentorship programme
- Mentor in K-12 Broadening Participation in Computing program at UIUC, Summer 2023.
(Guided a team of high-school students in successfully learning & conducting research on LLMs.)

Undergraduate Thesis Project

Security & Privacy of SCADA systems & Microgrids

IIT Delhi, India

Guides: Prof. Anupam Joshi & Prof. B.K. Panigrahi

Aug 2021 - Dec 2021

- Developed Multi-task data-driven real-time Intrusion detection system for digital twin of AC Microgrid (AUC: 0.85)